

PYQ for Software Enginnering

Compiled from CAT & End-Sem Questions

1 1 Marks Questions

1. Define generic software with example. [End 2026]
2. What do you understand by the term life cycle model of the software development? [End 2026]
3. Explain the need for SRS. [End 2026]
4. What is the difference between module coupling and module cohesion? [End 2026]
5. Distinguish between verification and validation. [End 2026]
6. List two advatanges disadvantages of LOC. [End 2026]
7. What are the objectives of testing? [End 2026]
8. Discuss some of the problems that are faced during maintenance of software. [End 2026]

2 5 Marks Questions

1. Consider the problem of railway reservation System and design the following:
 - (a) Problem Statement
 - (b) Use Case Diagram
 - (c) Use Cases

[CAT 2026]

2. Consider an example of grading a student in a university. The grading is done as given below:

Average Marks	Grade
90 - 100	Exemplary performance
75 - 89	Distinction
60 - 74	First Division
50 - 59	Second Division
0 - 49	Fail

The marks of any three subjects are considered for the calculation of average marks. Generate boundary value analysis test cases. Also create equivalence classes and generate test cases. [CAT 2026]

3 6 Marks Questions

1. What is Software Requirement Specifacation (SRS)? Explain the characteristics of a good Software Requirement Specification. [End 2026]

2. (a) You are asked to develop an accounting system for CUSB Gaya that replaces the existing system. Suggest the most appropriate software process development model that might be used as a basis for managing system development. Give proper justification.
(b) Under what circumstances is it beneficial to construct a prototype? Does construction of a prototype always increase the overall cost of software development?
[End 2026]
3. Explain the types of cohesion and coupling in the context of software design. [End 2026]
4. Explain why the best programmers do not always make the best software managers. You may find it helpful to base your answer on the software project management activities.
[End 2026]
5. Why is accurate estimation of the effort required for completing a project difficult? Briefly explain the different effort estimation methods that are available. [End 2026]
6. (a) Annual Change Traffic (ACT) for a software system is 13% per year. The development effort is 600 PMs. Compute estimate for Annual Maintenance Effort (AME). If the life span of the project is 10 years, what is the total effort of the project (using Boehm Model)?
(b) Suppose that a project was estimated to be 500 KLOC. Calculate the effort and development time for each of the three modes of Basic COCOMO model i.e., organic, semi-detached and embedded.
[End 2026]
- 7.

4 15 Marks Question

1. What is black box testing? Explain the different types of black box testing with example.
[End 2026]
2. (a) Consider a software project with 5 tasks T1 - T5. Duration of the 5 tasks (in days) are 15, 10, 12, 25 and 10 respectively. T2 and T4 can start when T1 is complete. T3 can start when T2 is complete. T5 can start when both T3 and T4 are complete. The project is to be completed in the shortest time.
 - i. Calculate the early time and late time for each event.
 - ii. Find the critical paths and the critical activities.
- (b) What is the purpose of the data flow diagrams? Explain by constructing a context flow diagram, level -1 DFD and level -2 DFD for a library management system?
[End 2026]
3. Write short notes on any two of the following:
 - (a) Different types of maintenance
 - (b) Path Testing
 - (c) Use Case Diagram

[End 2026]